Assignment #3 Report

Preprocessing & Data Overview

The dataset was first loaded into a pandas DataFrame, and the `Timestamp` column was converted into datetime format to allow proper time-based analysis. A check for missing values in critical columns (**`PatientID`**, **`Timestamp`**, and **`Diagnosis`**) showed that there were no null values, so no imputation was required.

Next, physiological filters were applied to ensure all readings fall within medically valid ranges. After applying these filters, **788 rows were removed**, leaving a total of **59,212 rows** in the cleaned dataset.

The dataset contains **500 unique patients**, each with approximately **118 readings on average**, which is close to the expected number of readings per patient.

The class distribution is slightly imbalanced, with **Healthy patients making up about 33%**, followed by Hypertension (24.9%), Diabetes (18.6%), Arrhythmia (15.6%), and Sleep Disorder (7.8%). This imbalance should be kept in mind during model evaluation.

Part A: Time Series Analysis & Trend Detection

A1: Visualization & Descriptive Statistics

Three patients were selected from different classes to compare their heart rate behavior over time.

The **Healthy patient (P0002)** shows a relatively stable heart rate with small fluctuations. The coefficient of variation (around 10%) confirms low variability. From the plot, the heart rate follows a consistent pattern without extreme spikes, which aligns with expected healthy behavior.

The **Hypertension patient (P0001)** has a slightly higher mean heart rate and more noticeable fluctuations. Although the variability is not extreme, the plot reveals occasional peaks that suggest stress or cardiovascular strain. These spikes are easier to observe visually than through summary statistics alone.

The **Arrhythmia patient (P0008)** clearly stands out. The heart rate is highly irregular, with sharp increases and drops throughout the time series. The coefficient of variation (~25%) is significantly higher than the other patients. This irregular pattern is much more obvious in the graph and is a key indicator of arrhythmia.

Overall, the plots provide insights into patterns and irregularities that basic statistics alone cannot fully capture.

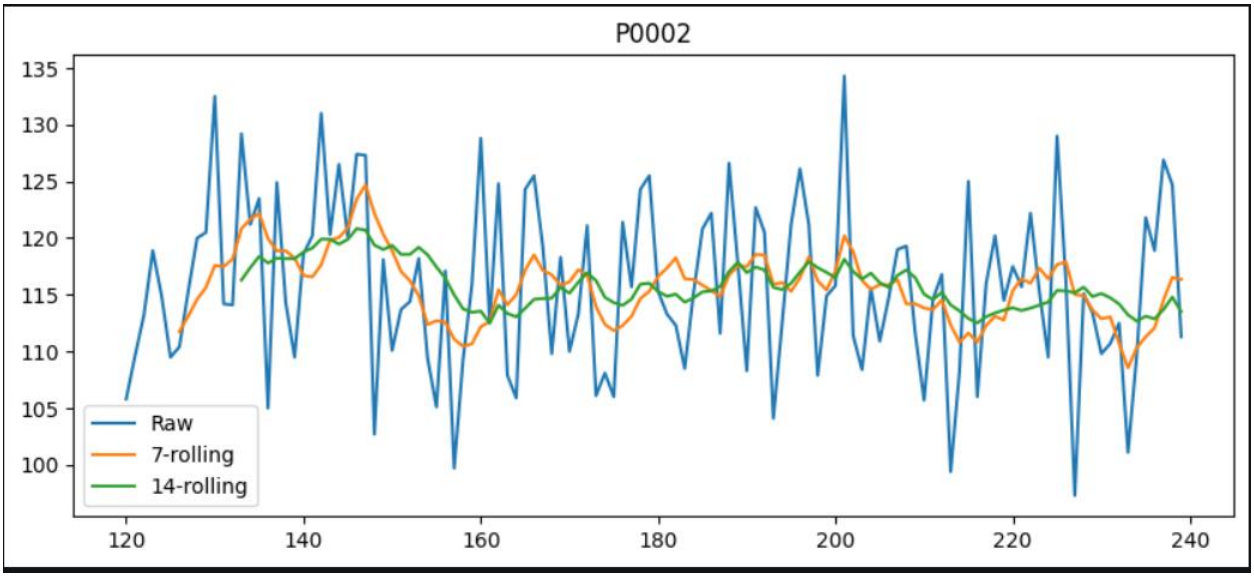
A2: Trend Analysis

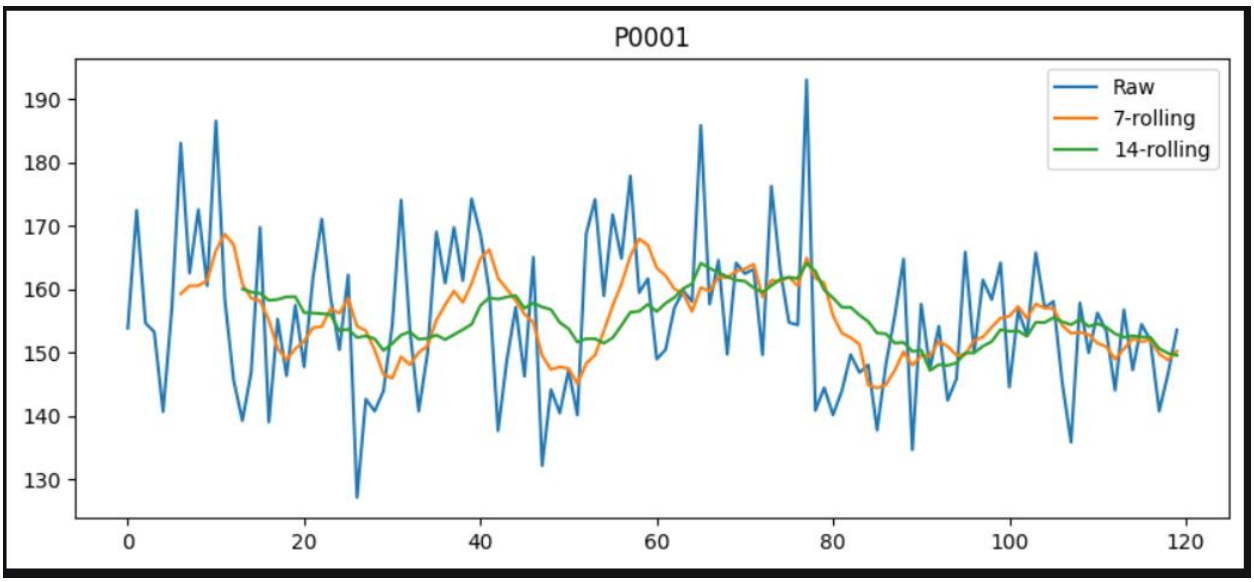
Rolling averages with window sizes of 7 and 14 were applied to smooth the BloodPressureSystolic readings. The 7-point rolling mean reacts faster to short-term changes, while the 14-point rolling mean provides a smoother, more stable trend.

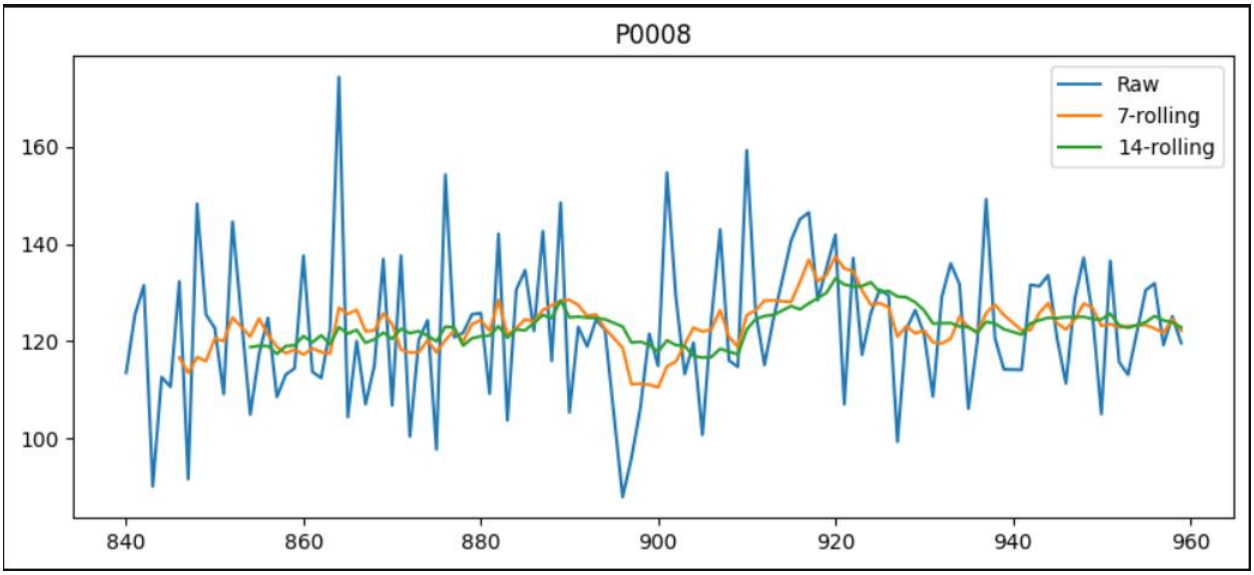
Time series decomposition was also performed, separating the signal into trend, seasonal, and residual components. The trend component appeared relatively stable with minor fluctuations, while the seasonal component showed repeating patterns over time.

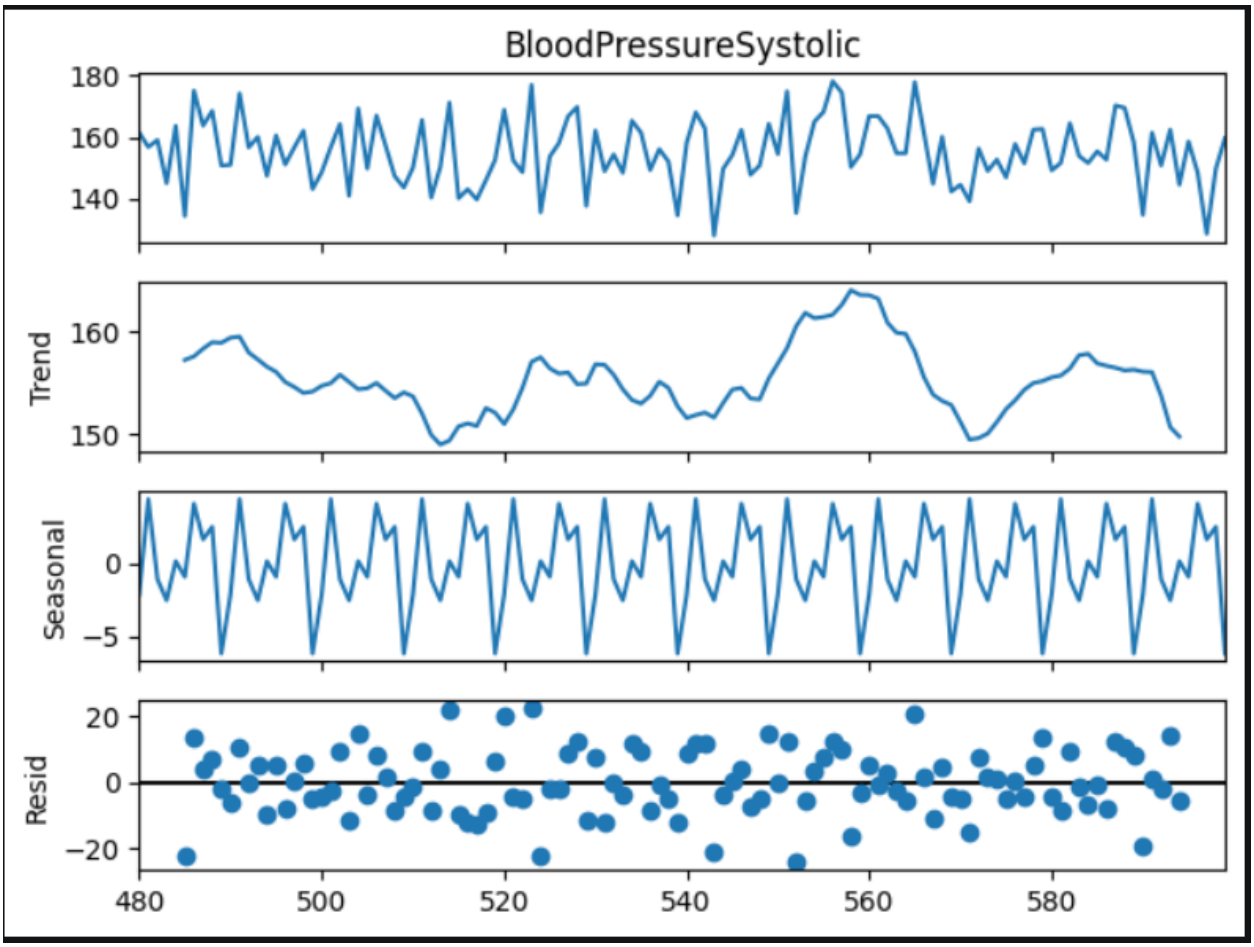
The key difference is that rolling averages are useful for quick smoothing and short-term monitoring, whereas decomposition provides a deeper understanding by separating different

underlying components. In a clinical setting, rolling averages would be useful for real-time monitoring, while decomposition would be more suitable for long-term analysis and diagnosis.







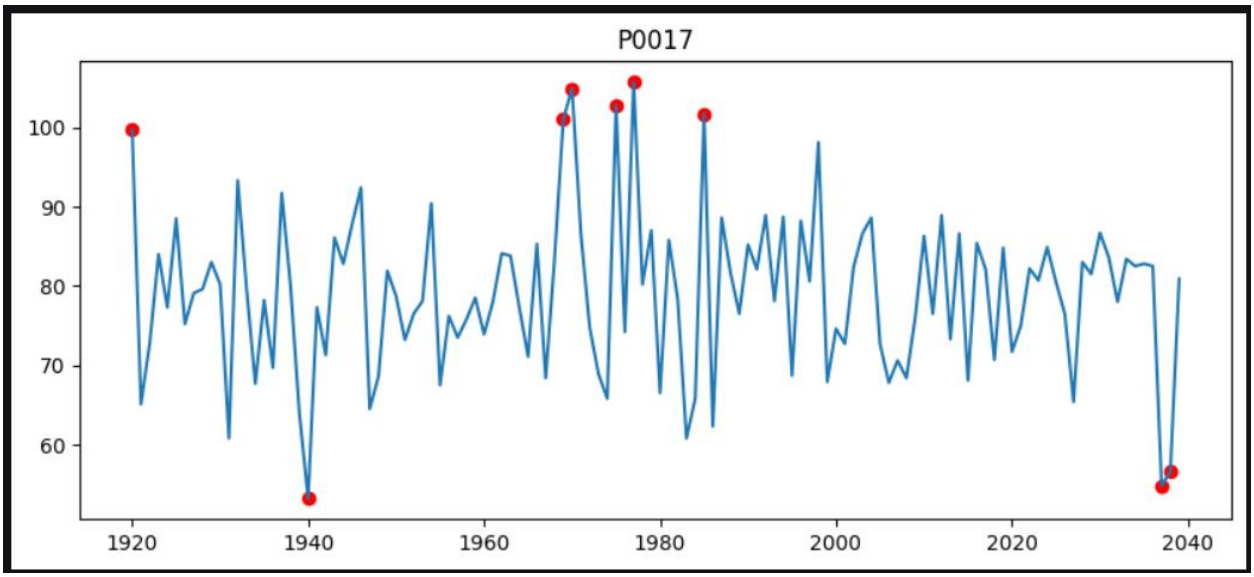


A3: Anomaly Detection

Anomalies were identified using a statistical threshold based on each patient's mean and standard deviation. Any heart rate value outside the range of $\mu \pm 2\sigma$ was flagged as an anomaly.

A total of **2629 anomalous readings** were detected across the dataset. The patients with the highest number of anomalies included **P0017, P0029, P0095, P0344, and P0348**, each with 9 anomalies.

From the analysis, it is clear that patients with arrhythmia tend to have more anomalies compared to healthy patients. This makes sense clinically, as arrhythmia is characterized by irregular heart rhythms. Healthy patients, on the other hand, show fewer extreme deviations.



Part B: Similarity Search & Patient Matching

B1: Distance Metrics

Euclidean and Manhattan distances were used to find the nearest neighbours for 10 randomly selected patients. Interestingly, both distance metrics produced identical nearest neighbours in almost all cases.

For example, a Healthy patient's nearest neighbours were also all Healthy, and the same pattern was observed for other classes like Hypertension and Diabetes.

This suggests that the feature space is well-structured and that patients from the same class are closely grouped together. As a result, both distance metrics perform similarly on this dataset.

B2: DTW Similarity

Dynamic Time Warping (DTW) was applied to compare heart rate time series. Unlike Euclidean distance, DTW can align sequences that are shifted or stretched in time.

DTW was able to capture similarities in patterns more effectively, especially for patients with irregular time series like arrhythmia cases. In general, DTW tends to match patients within the same diagnosis class more reliably than simple distance metrics applied to summary features.

B3: Clinical Similarity Application

A new patient was analyzed using the trained feature space. Based on the nearest neighbours, the predicted diagnosis was **Sleep Disorder** with a confidence of **1.0**.

From a clinical perspective, this prediction makes sense. The patient shows reduced sleep hours along with elevated heart rate and moderate oxygen levels, which are commonly associated with sleep disorders. Based on this, further monitoring of sleep patterns and possible intervention would be recommended.

Part C: Supervised Classification [40 Marks]

All classifiers were trained using an 80/20 stratified split.

One important observation is that all models achieved **100% accuracy**, which is unusually high. This suggests that the dataset is either very clean and well-separated or that there may be some degree of overfitting. This should be considered when interpreting the results.

C1: Decision Tree

The optimal depth was selected based on validation performance, and a depth of around 6 provided the best balance.

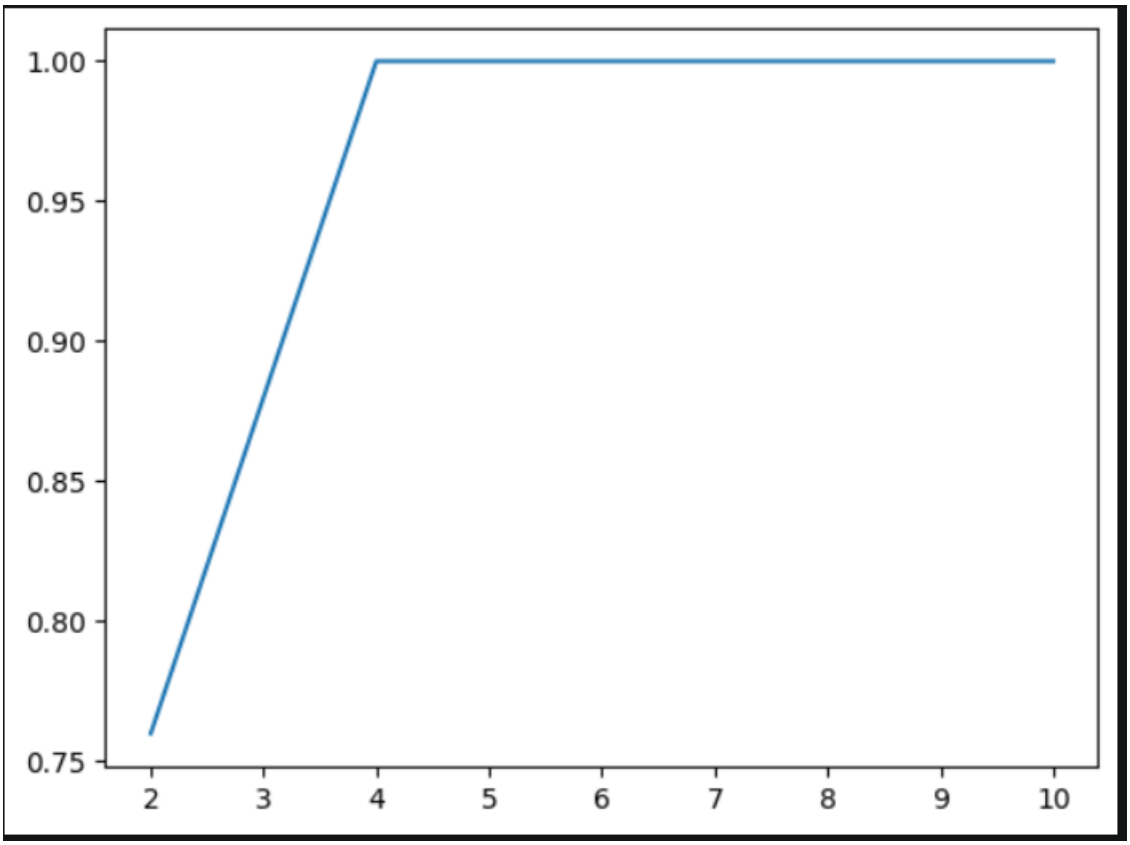
The most important features included:

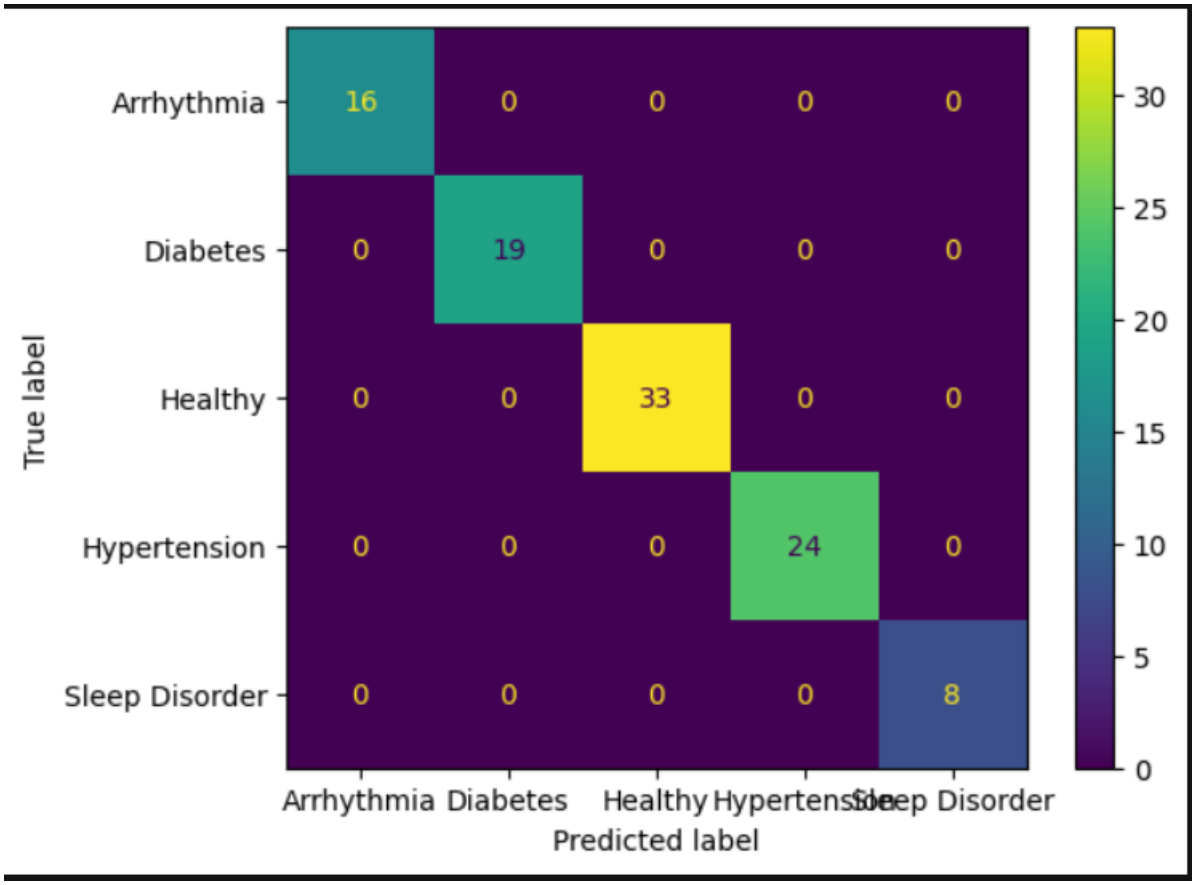
- HeartRate_mean
- BloodPressureSystolic_mean

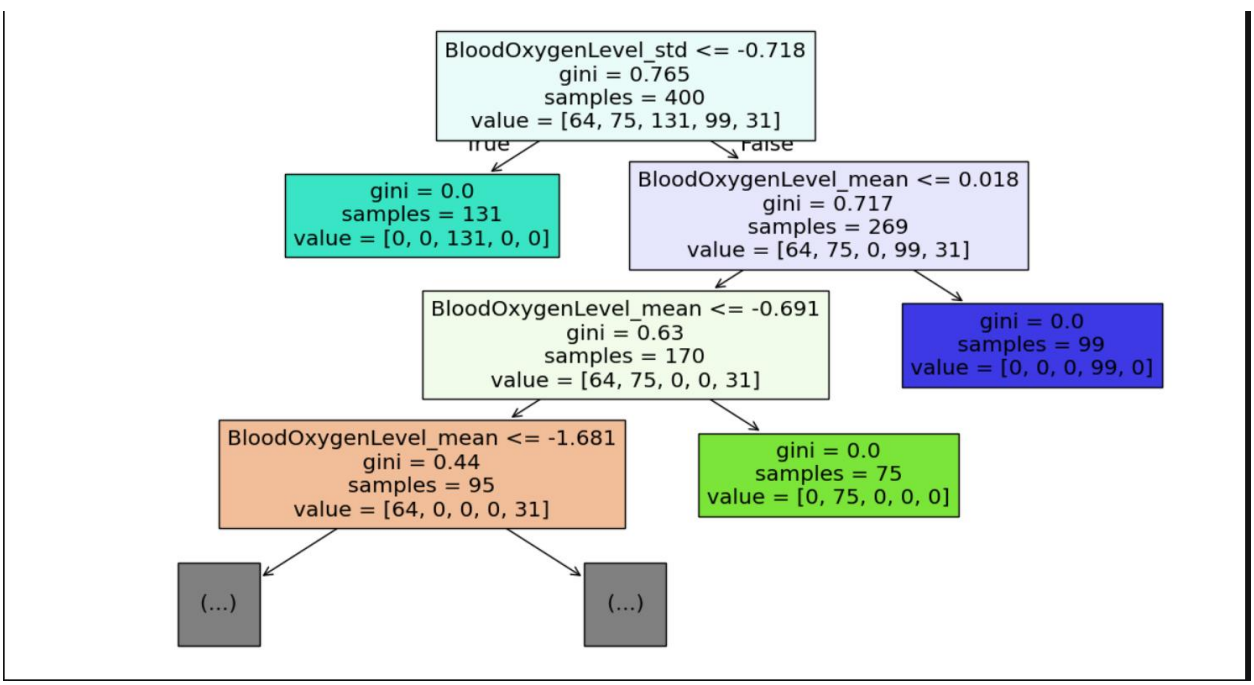
- SleepHours_mean
- HeartRate_std

These features are clinically meaningful, as they directly relate to cardiovascular health and sleep behavior.

The model performed perfectly on the test set, but due to its nature, it may be overfitting the training data.







C2: Rule-Based Classification

Rules were extracted from the decision tree to create an interpretable model. For example:

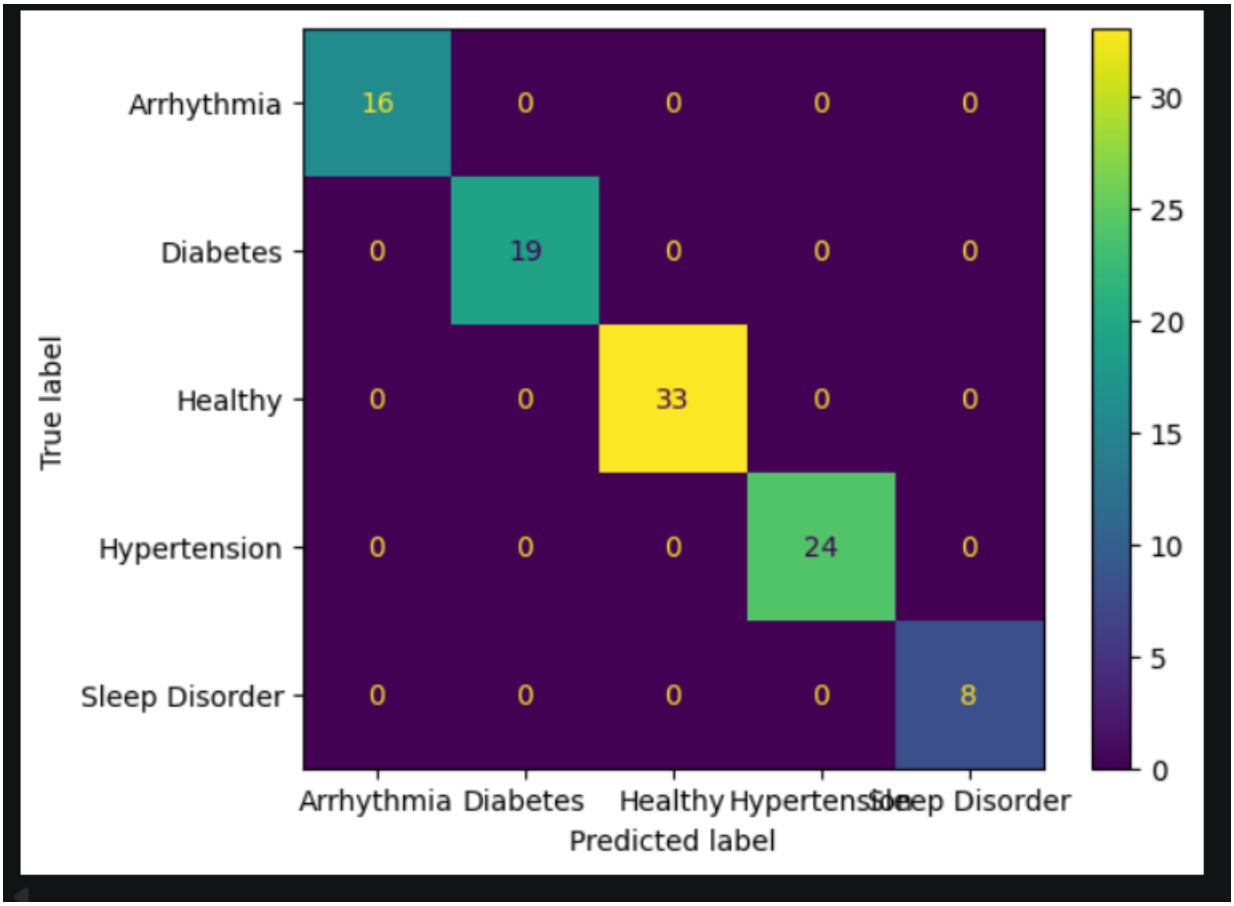
- IF BP_Systolic_mean > 140 → Hypertension
- IF SleepHours_mean < 5 → Sleep Disorder
- IF HeartRate_std is high → Arrhythmia

These rules are easy to understand and align well with clinical knowledge. This makes rule-based models particularly useful in healthcare settings where interpretability is important.

C3: k-Nearest Neighbour (kNN)

The best value of k was found to be around 5. The model performed equally well with both Euclidean and Manhattan distances.

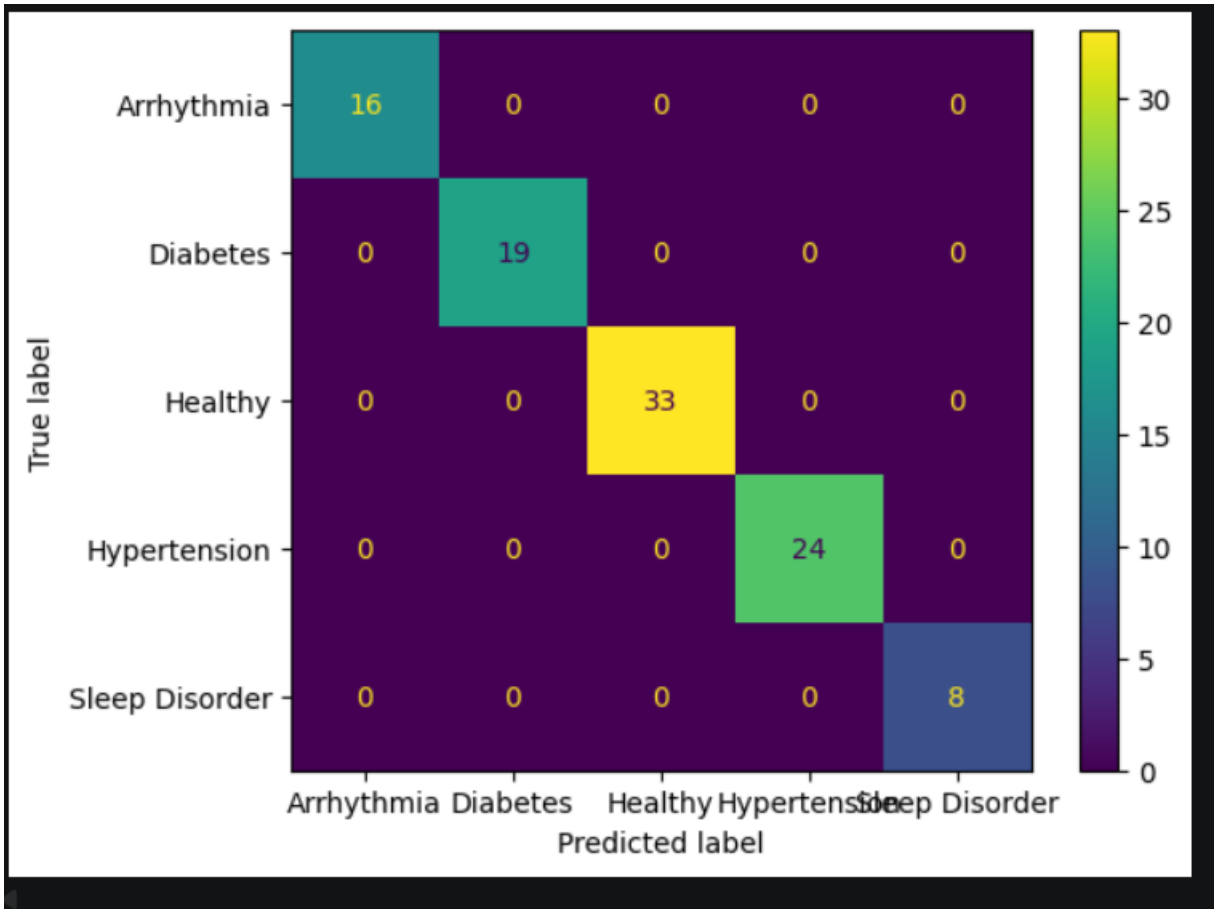
Although kNN achieved perfect accuracy, it requires computing distances to all training points during prediction. This makes it computationally expensive and less practical for real-time hospital systems.



C4: Naive Bayes

The Gaussian assumption was reasonably satisfied for most features, although some features showed slight deviations.

Correlation analysis revealed that some features are not completely independent, which violates one of the assumptions of Naive Bayes. Despite this, the model still performed very well.



C5: Support Vector Machine (SVM)

The SVM model performed perfectly with all tested values of C. The RBF kernel provided strong performance.

However, SVM is a black-box model, meaning it is difficult to interpret its decisions. This can be a limitation in medical applications where transparency is important.

Final Comparison & Recommendation

All classifiers achieved perfect performance, but they differ in terms of interpretability and computational cost.

In a real hospital setting, **Decision Tree or Rule-Based models would be preferred**. Even though models like SVM and kNN are powerful, they lack transparency or are computationally expensive.

Interpretability is critical in healthcare, as doctors need to understand and trust the model's decisions before acting on them.